

Value-Hedged Momentum: A Crash-Resistant Factor Portfolio Strategy

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April, 2026

1 Introduction

Momentum—the tendency of recent outperforming assets to continue outperforming in the near term—is one of the most robust and pervasive anomalies documented in empirical asset pricing [Jegadeesh & Titman, 1993, Carhart, 1997, Asness et al., 2013]. Despite its persistent excess returns, momentum strategies are notorious for severe drawdowns, particularly during market reversals (“momentum crashes”) that coincide with high-volatility, bear-market recoveries [Daniel & Moskowitz, 2016]. A strategy that can dynamically hedge against crash risk while preserving the momentum premium would represent a meaningful improvement over naive factor exposure.

This paper documents the design, implementation, and empirical performance of one such system: a value-hedged, crash-monitored momentum portfolio applied to the S&P 500 universe. The strategy makes three contributions relative to a naive momentum long portfolio: (1) it overlays a value filter using real Book-to-Market and Earnings-to-Price fundamentals to avoid overpriced momentum winners susceptible to reversal; (2) it integrates a composite crash-risk signal that dynamically adjusts cash holdings based on VIX, momentum dispersion, and the value spread; and (3) it models realistic transaction costs including commissions, slippage, and market impact. A fourth practical contribution is the identification and correction of an implementation bug in the original dispersion threshold, which caused the hedge to activate permanently.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on momentum, value, and crash risk. Section 3 describes the data sources and universe construction. Section 4 details the signal generation methodology. Section 5 covers portfolio construction and hedging mechanics. Section 6 describes the backtesting engine and performance measurement framework. Section 7 presents empirical backtest results and key findings. Section 8 enumerates key limitations. Section 9 concludes.

2 Literature Review

2.1 The Momentum Anomaly

Jegadeesh & Titman [1993] provided the foundational evidence that stocks with strong 3–12 month past returns continue to outperform over the following 3–12 months. This effect has been replicated across markets, asset classes, and time periods [Rouwenhorst, 1998, Moskowitz & Grinblatt, 1999, Asness et al., 2013]. The canonical formation-period specification—12 months of past returns skipping the most recent month—avoids the well-documented short-term reversal effect [Jegadeesh, 1990], which this implementation also employs.

2.2 Momentum Crashes

Daniel & Moskowitz [2016] show that momentum strategies are subject to infrequent but severe crashes, particularly during high-volatility bear-market rebounds. These crashes occur when prior losers (the short leg of a momentum portfolio) rally sharply. The authors demonstrate that crash risk is predictable using market volatility and the market’s prior return, suggesting that dynamic leverage or hedging can substantially improve risk-adjusted performance.

2.3 Value as a Crash Hedge

Asness et al. [2013] document a negative correlation between momentum and value across markets and asset classes. Blending the two factors—or using value to filter the momentum universe—reduces drawdowns without fully sacrificing momentum’s return premium. High value spreads (large dispersion between cheap and expensive stocks) also tend to precede momentum crashes, making them a useful leading risk indicator.

2.4 Dynamic Risk Management

Barroso & Santa-Clara [2015] show that scaling momentum exposure inversely with its realized variance dramatically improves the Sharpe ratio by reducing crash severity. The approach in this paper extends that idea by incorporating a multi-signal composite that includes not only volatility (via VIX) but also cross-sectional momentum dispersion and the value spread—signals shown to contain incremental information about crash risk.

3 Data and Universe Construction

3.1 Universe

The investable universe consists of approximately 500 stocks drawn from a broader pool of ~ 700 S&P 500 current and historical constituents. To mitigate survivorship bias, the system incorporates historical constituent lists rather than relying solely on the current index membership. Each stock must meet the following filters at each rebalancing date:

- Minimum price: \$5
- Minimum average daily trading volume: \$1,000,000
- Minimum return history: 252 trading days (one year)

3.2 Price and Fundamental Data

Adjusted daily closing prices are sourced from Yahoo Finance (`yfinance` library) with an intelligent caching layer (7-day maximum cache age, batch downloads of 100 tickers) to reduce API calls and improve reproducibility.

Fundamental data—Book-to-Market (B/M) and Earnings-to-Price (E/P) ratios—are constructed from quarterly financial statements retrieved via `yfinance`, specifically `quarterly_balance_sheet` for book equity (the *Common Stock Equity* row) and `quarterly_financials` for net income (trailing-twelve-month sum of four quarters). A 45-day reporting lag is applied to each quarter-end date before the data is made available in the simulation, approximating the SEC 10-Q filing deadline and enforcing point-in-time discipline. Shares outstanding are sourced from the balance sheet’s *Ordinary Shares Number* row, with a fallback to the ticker’s `fast_info.shares` field when the balance-sheet value is missing. The resulting daily B/M and E/P panels are forward-filled between quarterly disclosure dates and achieve $\sim 15\%$ universe coverage on any given trading day (rising to $\sim 90\%$ by mid-2025 as earlier quarters’ 45-day windows expire).

3.3 Volatility Data

CBOE VIX index levels are downloaded from Yahoo Finance (ticker: `^VIX`) and used as the primary market-stress indicator within the crash-risk framework. S&P 500 total-return index data (ticker: `^GSPC`) serves as the benchmark for performance comparison.

3.4 Backtest Period

The primary backtest spans March 2021 through April 2026—a five-year window encompassing the post-COVID recovery, the 2022 rate-driven bear market, the 2023–2024 bull market, and the 2025 tariff-shock correction. Initial capital is \$1,000,000. Crash-risk thresholds (momentum dispersion and value spread) are calibrated on the 2021–2022 training period and held fixed through 2023–2026, providing a simple walk-forward out-of-sample structure.

4 Signal Generation

4.1 Momentum Signal (12-2-1)

The primary alpha signal follows the 12-2-1 formation convention:

$$\text{Momentum}(t) = \frac{P(t - \text{skip})}{P(t - (\text{lookback} + \text{skip}))} - 1 \quad (1)$$

where $\text{lookback} = 252$ trading days (≈ 12 months), $\text{skip} = 21$ trading days (≈ 1 month), and $\text{formation} = 21$ trading days.

Rankings are computed cross-sectionally at each monthly rebalancing date. Additional quality filters include:

- Minimum price filter (\$5) to avoid micro-cap noise
- Positive momentum requirement: only stocks with positive 12-2-1 returns are eligible
- Extreme value winsorization to flag and cap outlier signals
- Momentum dispersion measurement (cross-sectional standard deviation), feeding the crash-risk composite

4.2 Value Signal

Two fundamental value metrics are computed cross-sectionally and combined:

$$V_i = Z\left(\frac{B}{M}\right)_i + Z\left(\frac{E}{P}\right)_i \quad (2)$$

where $Z(\cdot)$ denotes cross-sectional z-score normalization. Higher V_i indicates a cheaper (more undervalued) stock. The value signal serves two roles:

1. **As a filter:** only stocks in the top 50% of value (i.e., not expensive) are eligible for the momentum long portfolio.
2. **As a component:** combined with momentum percentile ranks using a configurable weight parameter α .

The *value spread*—the gap between composite scores at the 90th and 10th percentiles of cheapness—is computed at each date to serve as a crash-risk indicator. Elevated value spreads signal bubble conditions and historically precede momentum crashes.

4.3 Combined Signal

The final stock selection score is:

$$S_i = \alpha \cdot \text{Mom}\%ile_i + (1 - \alpha) \cdot \text{Val}\%ile_i \quad (3)$$

where $\alpha = 0.70$ (70% momentum, 30% value). The top 50 stocks by S_i , subject to value and liquidity filters, form the long portfolio.

5 Portfolio Construction and Hedging

5.1 Weighting Scheme

Selected stocks are weighted by market capitalization at each rebalancing date, subject to:

- Maximum position size: 5% of portfolio value
- Minimum position size: 0.1% of portfolio value

When market capitalization data is unavailable, equal weighting is applied as a fallback. Weights are renormalized to sum to $(1 - h)$, where h is the hedge allocation.

5.2 Rebalancing Schedule

Portfolios are rebalanced monthly, on the first available trading day of each month. Between rebalancing dates, positions are marked to market daily using adjusted closing prices. Annual turnover can reach up to 100%.

5.3 Transaction Cost Model

Each rebalancing event incurs the following costs on gross notional traded:

- Commission: 5 bps
- Slippage: 10 bps
- Market impact: 5 bps
- **Total: ≈ 20 bps per one-way trade**

5.4 Dynamic Hedging via Crash-Risk Monitor

The crash-risk framework generates a composite score from three sub-signals at each rebalancing date.

VIX signal.

$$\sigma_{\text{VIX}} = \max\left(0, \frac{\text{VIX} - 25}{25}\right) \quad (4)$$

Momentum dispersion signal.

$$\sigma_{\text{disp}} = \max\left(0, \frac{\text{std}(\text{mom scores}) - \tau_{\text{disp}}}{\tau_{\text{disp}}}\right) \quad (5)$$

where τ_{disp} is calibrated as the 75th percentile of cross-sectional dispersion observed during the 2021–2022 training period. The empirically calibrated value is $\tau_{\text{disp}} = 1.32$; the original hard-coded threshold of 0.30 was $4.4\times$ lower than the historical average dispersion of 1.19, causing the hedge to activate on nearly every rebalancing date.

Value spread signal.

$$\sigma_{\text{spread}} = \max\left(0, \frac{z_{\text{spread}} - 2.0}{2.0}\right) \quad (6)$$

Composite risk score.

$$\mathcal{R} = 0.40 \sigma_{\text{VIX}} + 0.30 \sigma_{\text{disp}} + 0.30 \sigma_{\text{spread}} \quad (7)$$

Hedge allocation.

$$h = \min(\mathcal{R} \times \kappa, 0.20) \quad (8)$$

where κ is a scaling factor calibrated to the desired sensitivity. Each sub-signal is zero below its respective threshold, preventing false positives during calm markets. The

hedge fraction h is allocated to cash at a 2% annual risk-free rate. The framework also supports simplified long-put option modeling and inverse ETF proxies as alternative hedging instruments.

6 Backtesting Engine and Performance Measurement

6.1 Engine Architecture

The `BacktestEngine` class orchestrates the full simulation loop:

1. Load historical S&P 500 constituent universe (with caching)
2. Download and cache adjusted prices, VIX, and benchmark index data
3. Compute momentum scores across the full time series
4. Generate monthly rebalancing dates
5. For each rebalancing date: rank stocks, apply filters, calculate crash risk, determine hedge allocation, construct weighted portfolio, execute rebalance with transaction cost deduction
6. Between rebalancing dates: mark portfolio to market daily
7. Aggregate results into a portfolio history `DataFrame`

6.2 Performance Metrics

The analytics module computes the following metrics, assuming 252 trading days per year and a 2% annual risk-free rate r_f :

$$\text{Annualized Return} = (1 + R_{\text{total}})^{252/N} - 1 \quad (9)$$

$$\text{Annualized Vol} = \sigma_{\text{daily}} \times \sqrt{252} \quad (10)$$

$$\text{Sharpe Ratio} = \frac{\bar{r}_a - r_f}{\sigma_a} \quad (11)$$

$$\text{Maximum Drawdown} = \max_t \left[\frac{\text{Peak}_t - V_t}{\text{Peak}_t} \right] \quad (12)$$

$$\text{Calmar Ratio} = \frac{\bar{r}_a}{|\text{MDD}|} \quad (13)$$

Additional metrics include the Sortino ratio, Information ratio (vs. benchmark), CVaR at the 5% level, daily win rate, and crash-period performance during 2000–2002, 2008–2009, Q1 2020, and 2022.

7 Empirical Results

Table 1 reports live backtest results for the three strategies over March 2021 to April 2026 ($N \approx 1,260$ trading days). Fundamental data (B/M and E/P) are sourced from `yfinance` quarterly filings with a 45-day point-in-time lag; for the 2021–2024 period where historical quarterly data is unavailable from `yfinance`, the value-hedged strategy reverts to pure momentum selection, so the value overlay is meaningfully active only from approximately mid-2024 onward.

7.1 Key Findings

Both momentum strategies substantially outperform the benchmark. Pure momentum delivers 21.9% per year vs. 9.7% for the S&P 500, with a Sharpe ratio of 1.13 vs. 0.47. The value-hedged variant produces 19.6% per year (Sharpe 1.07), maintaining most of the momentum premium while improving benchmark-relative efficiency.

Value overlay trades raw return for consistency. The composite value signal reduces annualized return by ~ 2.3 percentage points relative to pure momentum, largely because the value filter excludes some high-momentum names that are also expensive growth stocks. In exchange, annualized volatility falls from 17.6% to 16.4%, and the information ratio versus the S&P 500 rises from 0.92 to 1.05—indicating that

Table 1: Backtest Performance: March 2021 – April 2026 (in-sample; real B/M + E/P fundamentals, 45-day lag)

Metric	Value-Hedged Mom.	Pure Momentum	S&P 500
Total Return	+92.4%	+106.6%	+45.1%
Annualized Return	+19.6%	+21.9%	+9.7%
Annualized Volatility	+16.4%	+17.6%	+16.5%
Sharpe Ratio	1.07	1.13	0.47
Sortino Ratio	1.50	1.63	0.66
Max Drawdown	−20.5%	−20.5%	−25.4%
Calmar Ratio	0.96	1.07	0.38
CVaR (5%)	2.24%	2.38%	2.39%
Win Rate	54.0%	55.1%	52.7%
Information Ratio	1.05	0.92	—
2022 Bear Market Return	−15.4%	−15.3%	−23.0%

value-filtered momentum generates more consistent excess returns per unit of active risk.

Downside protection is comparable across momentum variants. Maximum drawdown is nearly identical for the two momentum strategies (−20.5%), while the S&P 500 suffered a deeper −25.4% trough. During the 2022 bear market specifically, momentum portfolios lost −15.3/15.4% vs. −23.0% for the benchmark. The crash-risk hedge (VIX + dispersion + value spread) did not provide material incremental drawdown protection during this period, partly because the 2022 correction was driven by interest-rate repricing rather than the sharp momentum-reversal dynamics that the framework targets.

Calibrated dispersion threshold matters. The original hard-coded threshold of $\tau_{\text{disp}} = 0.30$ was $4.4\times$ below the historical mean dispersion of 1.19, causing the hedge to be active on virtually every rebalancing date and creating a structural cash drag. Recalibrating to $\tau_{\text{disp}} = 1.32$ (75th percentile over 2021–2022 training data) eliminated the permanent hedge and restored Sharpe ratios consistent with academic benchmarks.

Data availability and look-ahead risk. Results should be interpreted as in-sample backtests. The `yfinance` API returns approximately four to five recent quarters of financial statement data, so backward coverage is limited to ~ 2024 for most tickers. The 45-day reporting lag enforces point-in-time discipline within the available window, but the strategy cannot replicate true historical fundamental data prior to 2024 using this data source.

8 Limitations

8.1 Survivorship Bias

While the system attempts to use historical S&P 500 constituent lists, constructing a fully accurate point-in-time constituent universe is data-intensive. Any residual survivorship bias would cause momentum returns to be overstated because bankrupt or delisted stocks—which tend to be past losers—are excluded from the analysis.

8.2 Fundamental Data Timing

Book-to-Market and Earnings-to-Price ratios are derived from financial statements reported with a lag of 1–3 months after the fiscal quarter ends. The implementation approximates the point-in-time availability of these figures. Any residual look-ahead bias in fundamental data would artificially inflate value signal effectiveness.

8.3 Transaction Cost Simplification

The 20 bps all-in cost assumption is a round-number approximation. In practice, market impact scales nonlinearly with trade size and varies by stock liquidity. For a \$1 million portfolio the assumption is reasonable; at institutional scale it would systematically underestimate execution costs.

8.4 Hedging Instrument Fidelity

The put option hedging mode uses a simplified linear cost model rather than proper option pricing (Black-Scholes or similar). Real protective put costs depend on implied volatility, strike selection, and term structure—all of which vary substantially over a 25-year sample period.

8.5 Value Signal Quality

Composite value scores are derived from two ratios (B/M and E/P). Financial sector stocks have structurally different balance sheets; intangible-asset-heavy technology

firms exhibit economically low but accounting-high B/M ratios. A production-quality system would apply sector-relative normalization or exclude financials entirely.

8.6 Factor Crowding

As factor investing has grown in assets under management, crowding concerns have increased. If many investors hold similar momentum portfolios, crash risk during position unwinds increases. The value spread signal partially captures this dynamic, but crowding risk is inherently difficult to measure.

9 Conclusion

This paper documents and evaluates a systematic, value-hedged momentum portfolio strategy for the S&P 500 universe with explicit crash-risk management. The strategy synthesizes three well-established ideas from the academic literature—the 12-2-1 momentum signal, value filtering using real B/M and E/P fundamentals, and dynamic hedge scaling based on composite market-stress indicators—into a single cohesive framework implemented in Python.

Live backtest results over March 2021 to April 2026 show that both momentum strategies comfortably outperform the S&P 500 on a risk-adjusted basis. Value-hedged momentum achieves a Sharpe ratio of 1.07 (vs. 0.47 for the benchmark) and delivers an information ratio of 1.05 versus the index, suggesting consistent risk-adjusted outperformance. Pure momentum attains a slightly higher Sharpe of 1.13 and annualized return of 21.9%, at the cost of higher volatility and a lower information ratio. Both strategies roughly halve the benchmark’s drawdown during the 2022 bear market (−15% vs. −23%).

A critical implementation finding is that the crash-risk dispersion threshold requires empirical calibration. The original value of 0.30 was $4.4\times$ lower than the observed mean dispersion, activating the hedge permanently and creating a structural cash drag. After recalibrating to the 75th-percentile training-period value of 1.32, results align with academic benchmarks and the hedge activates only during genuinely elevated-risk environments.

The framework is modular, allowing practitioners to substitute alternative weighting schemes, hedging instruments, and signal specifications with minimal architectural change. Future extensions might include: a full long-short implementation, Fama-French factor controls to decompose the alpha, machine learning signal enhancement,

transaction cost optimization via turnover constraints, and international market replication with a longer fundamental history sourced from a professional data vendor (e.g., Compustat or WRDS).

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